

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

**PRAGYA G8 – ACHIEVERS | MATHEMATICS
LOGARITHMS | LEVEL 1***Straightforward – one idea, one step.*1. The value of $\log_{0.01} 1000 + \log_{0.1} 0.0001$ is

(A) $-\frac{5}{2}$

(B) $\frac{5}{2}$

(C) -10

(D) -2

2. If $\log_2(\log_3(\log_4 x)) = 0$, $\log_3(\log_4(\log_2 y)) = 0$ and $\log_4(\log_2(\log_3 z)) = 0$, then $x + y + z$ is

(A) 58

(B) 50

(C) 89

(D) 105

3. If $\log_2(4 + \log_3 x) = 3$, then the sum of the digits of x is

(A) 6

(B) 18

(C) 9

(D) 3

4. The value of $\log_3(\log_2(\log_{\sqrt{3}} 81))$ is

(A) 1

(B) 0

(C) 3

(D) 2

5. $\log_3 \left[\log_2^2 \left(\frac{1}{2} \right) + 6 \log_2 \sqrt{2} + 5 \right] =$

(A) 9

(B) 3

(C) 2

(D) 1

6. $\log \frac{75}{16} - 2 \log \frac{5}{9} + \log \frac{32}{243} =$

(A) 1

(B) 0

(C) $\log 3$

(D) $\log 2$

7. $x^{\ln y - \ln z} \cdot y^{\ln z - \ln x} \cdot z^{\ln x - \ln y} =$

(A) 1

(B) 0

(C) xyz

(D) $x + y + z$

8. $\frac{\log_2 32}{\log_3 \sqrt{243}} =$

(A) $\frac{5}{2}$

(B) 2

(C) $\frac{1}{2}$

(D) 5

9. $\frac{2 \log 6}{\log 12 + \log 3} =$

(A) $\frac{1}{2}$

(B) 2

(C) $\log 2$

(D) 1

10. $\log_{1/4} \left[\left(\frac{1}{16} \right)^{-2} \right] =$

(A) 2

(B) 4

(C) -4

(D) -2

11. $\frac{\log_5 16 - \log_5 4}{\log_5 128} =$

(A) $\frac{2}{7}$

(B) $\frac{1}{2}$

(C) $\frac{4}{7}$

(D) $\frac{7}{4}$

12. If $\log x + \log 5 = \log x^2 - \log 14$ (base 10), then x equals

(A) 0

(B) 70^2

(C) 2^{70}

(D) 70

13. If $a^4 \cdot b^5 = 1$, then $\log_a (a^5 b^4)$ equals

(A) 5

(B) 4

(C) $\frac{8}{5}$

(D) $\frac{9}{5}$

14. Given $\log_{10} 2 = a$ and $\log_{10} 3 = b$. If $3^{x+2} = 45$, then x in terms of a and b is

(A) $\frac{1+a}{b}$

(B) $\frac{1-a}{b}$

(C) $\frac{b}{1-a}$

(D) $\frac{a-1}{b}$

15. If $\log_y x + \log_x y = 7$, then $(\log_y x)^2 + (\log_x y)^2$ is
(A) 47 (B) 45
(C) 49 (D) 43
16. The antilogarithm of $0.\bar{6}$ to the base 27 is
(A) 3 (B) 27
(C) 9 (D) 18
17. The integral part of $\log_2 2008$ is
(A) 11 (B) 10
(C) 12 (D) 9
18. The value of b satisfying $\log_e 2 \cdot \log_b 625 = \log_{10} 16 \cdot \log_e 10$ is
(A) 25 (B) 4
(C) 8 (D) 5
19. $\frac{1}{1 + \log_b a + \log_b c} + \frac{1}{1 + \log_c a + \log_c b} + \frac{1}{1 + \log_a b + \log_a c} =$
(A) 1 (B) 0
(C) abc (D) $\log(abc)$
20. The sum of all the solutions of the equation $2 \log x - \log(2x - 75) = 2$ is
(A) 200 (B) 30
(C) 75 (D) 350
21. Let $x = 2^{\log 3}$ and $y = 3^{\log 2}$, where the base of the logarithm is 10. Then which one of the following holds good?
(A) $2y < x$ (B) $2x < y$
(C) $3x = 2y$ (D) $y = x$
22. If $4^{\log_9 3} + 9^{\log_2 4} = 10^{\log_x 83}$ ($x \in \mathbb{R}$), then x is
(A) 83 (B) 10
(C) 8 (D) $\frac{1}{10}$
23. If $\log_3 x = p$ and $\log_7 x = q$, which of the following yields $\log_{21} x$?

(A) $\frac{1}{p^{-1} + q^{-1}}$

(B) $\frac{1}{p + q}$

(C) $\frac{pq}{p^{-1} + q^{-1}}$

(D) pq

24. When the repeating decimal $0.363636\ldots$ is written as a fraction in lowest terms, the sum of its numerator and denominator is

(A) 20

(B) 15

(C) 8

(D) 16

25. The value of $7^{\log_{28} 112} \cdot 4^{\log_{28} 4}$ is

(A) 112

(B) 12

(C) 28

(D) 14